University of Mysore

MYSORE UNIVERSITY SCHOOL OF ENGINEERING

Manasagangotri campus, Mysuru-570006

(Approved by AICTE, New Delhi)

A Innovation and Design Thinking (21IDT18) Project Report

On

"Smart Rail Track Monitoring system"

Submitted By

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CERTIFICATE

This is to certify that the ***Innovation and Design Thinking Project (21IDT18)*** entitled ***"Smart Rail Track Monitoring System "*** is a bonafide work carried out by **Lasyapriya A. P ,I Semester, C Section ,** a student of **Mysore University School of Engineering in** partial fulfillment for the award of **Bachelor of Engineering** in **AI&ML of the University of Mysore, Mysuru**. It is certified that all corrections/suggestions indicated for have been executed by the above mentioned candidate.

Signature of Head of the Dept.(I/C)

Dr.Ananthapadmanabha.T
Director, MUSE

Innovation and Design Thinking (21IDT18) Project work Evaluation			
Sl.No	**Examiners**		**Date**
	Name	**Signature**	
1.			
2.			

DECLARATION

I, **Lasyapriya A. P** student of **I** semester**, C** Section, Department of **AI&ML**, Mysore University School of Engineering, Mysuru declare that the ***Innovation and Design Thinking Project (21IDT18) entitled "Smart Rail Track Monitoring System***", has been duly executed by me under incharge Faculty members.

The IDT project report of above entitled is submitted by me in the view of partial fulfillment of the requirement for the award of Bachelor of Engineering degree in Department of **Artificial Intelligence and Machine Learning** by University of Mysore.

Date: 25/ 11 /2023 **Lasyapriya A. P**

Place: Mysuru **Reg.No.**

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Abstract

The "Rail Track Monitoring System" project aims to revolutionize the maintenance and safety of railway infrastructure through the implementation of an advanced autonomous monitoring solution. The project focuses on developing a specialized rover equipped with state-of-the-art sensors and cameras to conduct real-time inspections of railway tracks. The rover's autonomous capabilities enable it to navigate through complex terrains and structures, providing comprehensive data on track geometry, alignment, and potential defects. The successful deployment of the Rail Track Monitoring System promises to enhance safety, reduce downtime, and optimize maintenance practices in railway operations. This project contributes to the ongoing evolution of railway infrastructure management, aligning with the industry's shift towards autonomous and technology-driven solutions.

Keywords: Monitoring, autonomous, real-time.

INTRODUCTION

Railway infrastructure networks are among the most important transportation assets and have served the worldwide growth of industry and civilization for hundreds of years. Railway is form of transportation which involves a large network of rail tracks all over the country. Trains are the moving bodies along these tracks to transport goods or people. As this system is totally dependent on the network of tracks , a small damage or obstacle on the way could cost lives of hundreds and also economical loss to the country.

To avoid this huge loss many alternative ideas have been implemented in the railway system. Technologies such as Trackside monitoring system, Anti-collision system, automated braking systems etc have been developed and adapted in most of the trains . These technology may prevent damage upto some extent, but for better safety measures we need a system like Rail track Monitoring Rover.

The Smart Rail Track Monitoring System Rover is an innovative project designed to enhance the efficiency and safety of railway infrastructure. This rover employs advanced technologies such as sensors, cameras, GPS tracking, faster and accurate communication and data analytics to monitor rail tracks in real-time, detect anomalies, and provide timely information to railway authorities for preventive maintenance. The primary goal of the rail track monitoring system is to proactively identify and address issues, thereby minimizing the risk of accidents and optimizing the overall performance of the railway network.

This report aims to provide an idea of how a rover can be used to prevent huge destruction. This also explain about working, technical idea, principle, implementation and practical applications of the rover. This is not just a rover but a rover with drone, simply means a flying rover. This kind of hybrid system combines the capabilities of both aerial and ground-based vehicles, providing a versatile platform for various applications. The Smart Rail Track Monitoring System by drone represents a cutting-edge paradigm shift in the realm of railway infrastructure management. In response to the critical need for more efficient and proactive monitoring of rail tracks, this project introduces a comprehensive solution that leverages the capabilities of unmanned aerial vehicles (UAVs) to revolutionize the traditional inspection processes.

By evaluating current Technology, developments, and practical perspectives, this report will provide a comprehensive overview of the opportunities and challenges surrounding this system. It will also emphasize the importance of further technological advancements, and widespread adoption of this innovative solution to combat Rail accidents and promote safer transportation in future.

Literature Review

[1] SarathChandran.P, Karthika.M.

The proposed system is an enhanced technique for monitoring the obstacle which uses arduino microcontroller, ultrasonic sensor, and radar module. The radar will obtain the distance from the obstacle and ultrasonic sensor will ensure to avert the accidents that may occur by the collision between the train and obstacle.

[2] Rahul Singh, Leena Sharma, Vandana Singh , Vivek Kr. Singh.

International Journal of Engineering Research & Technology Proposed a system which was able to detect the crack by the help of ultrasonic sensors, GPS and GSM module once the crack is detected the message along with the location is send to the officials.

[3] J. Trehag, P. Handel, and M. Ogren.

Onboard estimation and classification of a railroad curvature IEEE Trans. Instrum.The aim is to determine where the transition curves, straight tracks and circular curves are located along the railroad. For this purpose, the railroad curvature measurements are used. The method is based on a Double Exponential Smoothing algorithm (DES). The DES algorithm is described with the aid of linear filters, where the outputs are the estimate of the curvature and its rate of change.

[4] A. K. Shrivastava, A. Verma and S. P. Singh.

The system is implementable in the robotic sewer blockage detection system. The distance of the blockage from a specified entry point in the sewer pipeline can be calculated by adding travelled distance by the robotic vehicle and the distance of the blockage from the robotic vehicle. The accuracy of distance of blockage will be sufficient for normal practical uses. The system can be easily implemented in other devices and systems requiring the measurement of distance of an object or an obstacle from stationary or moving observation point where the ultrasonic sensor will be located.

PROBLEM STATEMENT

1.Broken Rail Tracks: The failure of a rail due to defects, wear and tear, or inadequate maintenance.



Fig no: 3.1 Damaged Rail Tracks

2.Debris : Foreign objects on the tracks, such as fallen trees, rocks, or debris from previous accidents, can obstruct the path of trains.



Fig no: 3.2 Unwanted objects on rail track

3.Inadequate Maintenance: Lack of proper maintenance and inspection of tracks, signals, and rolling stock can result in deteriorating conditions.

4.Insufficient Inspection Procedures : Inadequate inspection protocols or failure to identify and address potential issues during routine checks.

5.Track Geometry Issues: Poorly aligned tracks, irregularities in track gauge, or improper maintenance of track geometry can lead to accidents.



Fig no: 3.3 Wrong alignment of tracks

6.Criminal Activities: Deliberate actions such as vandalism, sabotage, or unauthorized access to railway infrastructure can lead to accidents.



Fig no: 3.4 Stones arranged on tracks

7.Broken Switches: Malfunctions or breakages in railroad switches can lead to train derailments at junctions. If a switch is not set correctly, it can direct a train onto the wrong

8.Construction Work: Inadequate safety measures during construction or maintenance work near the tracks can pose risks.

9.Collapse of bridges or tunnels: Structural issues with bridges and viaducts can lead to collapses or other accidents.

METHODOLOGY

There are many types of technologies used for the maintenance of the railway tracks currently, some of these inspection machines are Push-Trolley, Road–Rail Vehicle, Specialist Train, Robotic Inspection Technologies etc,. Keeping these technologies as the basic structure of the rover, and alter the structure as an unmanned aerial and ground vehicle. This could be capable of inspecting the track as well as communicating with the control room of railway authorities with quick and accurate information.

i)This is how it prevents Rail accidents:

Basically this rover is designed in such a way that it moves on the rail track ,having the same track gauge as a train. Now this rover is mounted with a drone body, that can actually lift the complete rover whenever necessary. This system is equipped with 360 degree cameras, IR sensors, Ultrasonic sensors, signal receivers, embedded digital communication system and so on. When it moves along the track for inspection and finds out some damage or any obstacle , then immediately the message is sent to the control room of the respective nearby Railway station. This helps the authorities to inform the loco-pilots of any train passing on through that particular track. 360 degree cameras helps for real time monitoring the track, without any physical approach near the track. It simply means that , live inspection can be monitored or viewed from the respective station through systems. Each rover is programmed to continuously monitor the track up to certain distance limited from it's respective station. Whatever may be the circumstances or issues on the track, the rover would be the first to face the issue before the train, and thus saves the train from accidents. As the rover and the train are on the same track there may be a chance of collision between them. But the as the rover is equipped with a drone attached on it, which prevents it from collision. Using the motion sensors , as soon as the train approaches the whole system is raised above , higher than the train. This allows the train to pass without any collision. As soon as the train moves, the rover slowly lands on the track again and continues to monitor the track.

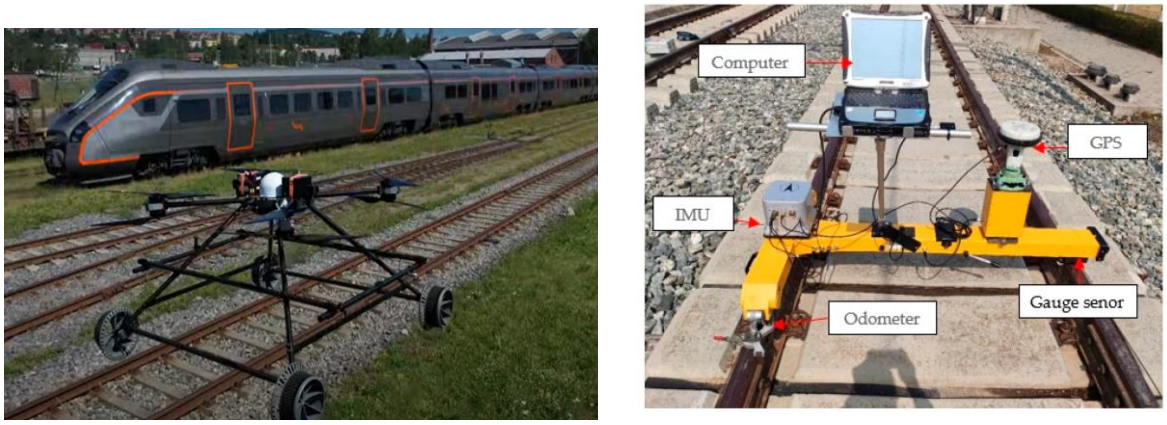


Fig no: 4.1 Inspection Rovers on rail track

ii) Basic Components of The Rover:

This specialised system consists of various components , which all together make the rover function properly and more advanced. Some of those components are as follows.

- Lithium-Ion batteries
- GPS Module
- Electronic speed controller
- Motors and Propellers
- Radio Transmission and Reciever
- Flight controller
- IR Sensors
- Ultrasonic Sensors
- 360 degree Cameras
- Motion Sensors
- Onboard Computer



Fig no: 4.2 Monitoring system on the track

1.Lithium-ion Batteries:

Lithium-ion (Li-ion) batteries are a type of rechargeable battery that has become ubiquitous in various electronic devices due to their high energy density, lightweight design, and long cycle life. lithium-ion batteries play a crucial role in powering the portable electronic devices that have become integral to modern life. Their combination of high energy density, rechargeability, and relatively low self-discharge makes them a popular choice across a broad spectrum of applications.



Fig no: 4.3 Lithium-ion Batteries

Method of recharging: Hybrid recharging technology refers to innovative approaches that combine multiple methods to recharge devices or systems. In the context of batteries and electric vehicles, it often involves the integration of two or more charging sources, such as traditional electrical outlets and renewable energy inputs like solar panels or regenerative braking. This hybrid approach aims to enhance energy efficiency, reduce reliance on conventional power sources, and promote sustainability in various applications, particularly in the realm of electric transportation and portable electronic devices

3.GPS Module:

The Global Positioning System (GPS) is a satellite-based navigation system that enables precise location determination anywhere on Earth. GPS relies on a network of satellites orbiting the Earth, with each satellite transmitting signals that allow a GPS receiver to calculate its precise three-dimensional position through trilateration. The accuracy and reliability of GPS have made it an essential technology for navigation, mapping, and a wide range of applications across various industries. This GPS enables the rover to identify and send message about where exactly the damage has occurred.

3.Electronic speed controller:

An Electronic Speed Controller (ESC) is a crucial component in the world of electric motors, commonly used in devices like drones, electric vehicles, and radio-controlled (RC) vehicles. Its primary function is to regulate the speed of an electric motor by adjusting the electrical current flowing to it. Electronic Speed Controller serves as the intermediary between the control system and the electric motor, regulating the motor's speed and direction based on input signals. Its versatility and efficiency make it a critical component in various electronic systems, contributing to the precise control of motors in a wide range of applications. Speed and acceleration of the rover is controlled by the Esc.

4.Motors and Propellers:

Motors provide the mechanical power necessary for drone flight, while propellers play a crucial role in converting that power into lift and propulsion, allowing the drone to move and maneuver in the air. The combination of efficient motors and well-designed propellers is essential for achieving optimal drone performance. Propellers are rotating blades that create a flow of air, generating lift and propulsion for the drone.





Fig no: 4.4 Motors and Propellers

5.Radio Transmission and Reciever:

Radio transmission and receivers are fundamental components in wireless communication systems, enabling the exchange of information over the airwaves. radio transmission involves the generation and propagation of modulated electromagnetic waves, while radio receivers capture, demodulate, and process these waves to extract the original information. This technology forms the basis of numerous wireless communication systems that play a vital role in modern telecommunications.

6.Flight controller:

A flight controller is a crucial component in the operation of drones, providing stability, control, and navigation during flight. A flight controller is a central component that plays a crucial role in the stability, control, and functionality of a drone. It integrates sensor data, user inputs, and sometimes GPS information to enable precise and responsive flight, making it a key technology for both recreational and professional drone applications.

7.Sensors:

Infrared (IR) sensors are devices that detect and measure infrared radiation, which is electromagnetic radiation with longer wavelengths than visible light. IR sensors play a versatile role in sensing and detecting infrared radiation, finding applications in proximity sensing, motion detection, temperature measurement, and communication systems. Their non-contact

nature and efficiency make them valuable components in a wide range of electronic devices and industrial processes.Ultrasonic sensors are devices that use sound waves of frequencies higher than the human audible range to measure distance or detect objects. ultrasonic sensors play a crucial role in distance measurement and object detection by utilizing high-frequency sound waves. Their non-contact nature, simplicity, and reliability make them valuable components in a wide range of industries and applications.

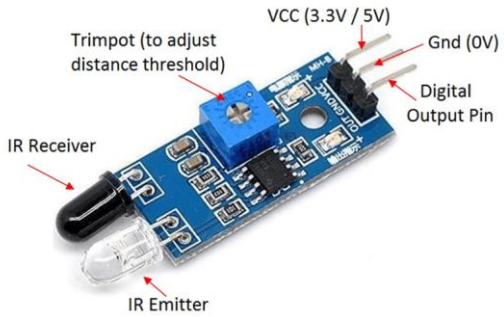


Fig no: 4.5 IR Sensors

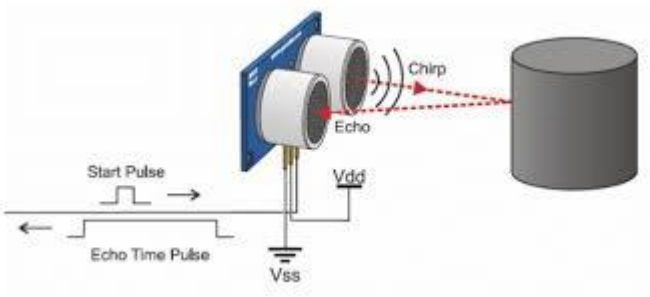


Fig no: 4.6 Ultrasonic sensors

8. 360 degree Cameras:

360-degree cameras, also known as omnidirectional cameras or VR cameras, capture images and videos in a full spherical view, covering all directions simultaneously. 360-degree cameras revolutionize content creation by providing a comprehensive view of the surroundings. They cater to a range of applications, offering users an immersive and interactive experience in the ever-expanding realm of virtual and augmented reality.



Fig no: 4.7 360 degree cameras

9.Motion Sensors:

Motion sensors work by sensing changes in their field, such as infrared radiation, sound waves, or microwaves. When motion is detected, the sensor responds by activating a connected system or device. Motion sensors are devices that detect movement in their surroundings and trigger a response. motion sensors play a pivotal role in various applications, contributing to energy efficiency, security, and the automation of processes in both residential and commercial settings.These sensors help the rover to lift above when train is about to approach.

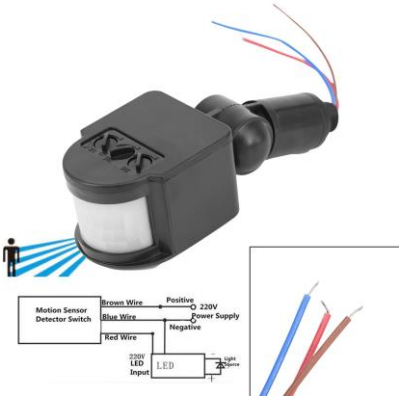


Fig no: 4.8 Motion Sensors

10. Onboard Computer:

An onboard computer refers to a computing device integrated into a larger system or vehicle, designed to perform specific tasks or control various functions. an onboard computer is a specialized computing device integrated into a larger system, tasked with processing data, making decisions, and controlling the functions of the overall system it is a part of.

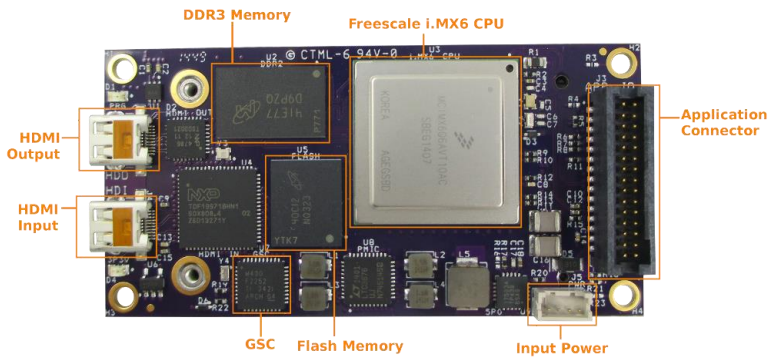


Fig no: 4.9 Onboard Computer Control board

ADVANTAGES

1.Safety Improvements: Inspection rovers can conduct thorough and frequent inspections of railway tracks without exposing human inspectors to potentially hazardous environments, such as busy rail lines or remote areas.

2. Efficient Regular Inspection: Rovers can operate autonomously or be remotely controlled, allowing for more frequent and consistent inspections of railway tracks. Regular inspections help identify and address issues promptly, reducing the risk of accidents and enhancing overall safety.

3. Early Detection Of Defects : Inspection rovers equipped with sensors and cameras can detect early signs of track defects, such as cracks, misalignments, or wear. Early detection allows for timely maintenance or repairs, preventing more significant issues that could disrupt train operations.

4.Cost Effective Maintenance: By identifying and addressing track issues proactively, inspection rovers contribute to cost-effective maintenance. Timely repairs and preventive measures can extend the lifespan of railway infrastructure and reduce the overall maintenance costs.

5. Data collection and analysis: Rovers equipped with advanced sensors can collect detailed data about the condition of tracks, including track geometry, alignment, and wear. This data can be analyzed to gain insights into the health of the railway infrastructure, informing maintenance strategies and decisions.

6. Remote Accessibility: Inspection rovers can access challenging or remote locations that may be difficult or unsafe for human inspectors. This includes areas with steep terrain, tunnels, or locations with limited visibility.

7.Minimized Disruptions to train Operations: Traditional manual inspections often require track closures, leading to disruptions in train operations. Inspection rovers can operate without interrupting regular train schedules, minimizing downtime and improving overall railway efficiency.

8. Real-Time Monitoring : Some inspection rovers provide real-time monitoring capabilities, allowing operators to receive immediate feedback on the condition of the tracks. This enables rapid response to emerging issues and reduces the likelihood of service interruptions.

9.Customization and Adaptability: Inspection rovers can be customized with various sensors and technologies based on the specific needs of the railway network. This adaptability ensures that the rovers are equipped to address the unique challenges of different rail environments.

10. Environmental Impact : By enabling targeted and efficient maintenance, inspection rovers contribute to reducing the environmental impact associated with railway infrastructure maintenance activities, such as the use of heavy machinery. Reduces pollution as heavy machines which run by combustion can be avoided.

11. Faster communication: Rover enables the station authorities to know the about the defect and inform the same to loco-pilots with least amount of time consumed. Accidents can be prevented much more efficiently.

DISADVANTAGES

1.Limited Mobility In certain Areas: Inspection rovers may face challenges in navigating complex or confined spaces, such as tunnels, bridges, or areas with dense vegetation. This limitation could result in incomplete inspections of critical sections of the railway.

2.Dependency On Technology: Rovers heavily depend on technology, including sensors and communication systems. Technical failures or malfunctions may occur, leading to disruptions in inspections and potential delays in identifying track defects.

3.Initial Cost and Investment: The upfront cost of acquiring and implementing inspection rovers, including the necessary technology and infrastructure, can be significant. This initial investment may pose financial challenges for some rail operators.

4.Maintenance and Keep-up: Like any mechanical or electronic system, inspection rovers require regular maintenance to ensure proper functioning. Maintenance costs, including software updates and hardware repairs, can contribute to the overall operational expenses.

5.Weather and Environmental Conditions: Harsh weather conditions, such as heavy rain, snow, or extreme temperatures, can affect the performance of inspection rovers. Adverse environmental conditions may limit the effectiveness of the technology.

Given that inspection rovers rely on communication systems and may store sensitive data, there are potential security concerns related to data breaches, unauthorized access, or cyberattacks.

Future Scope

Continued use of automatic inspection rovers for railways, including track inspection, bridge monitoring, and assessing the overall health of railway infrastructure. This can contribute to enhanced safety, reduced downtime, and optimized maintenance strategies. Application in the inspection of various infrastructures such as bridges, tunnels, and dams. Automatic inspection rovers can navigate through complex structures, providing detailed assessments of integrity and identifying maintenance needs. Inspection rovers can be deployed in industrial environments for monitoring manufacturing processes, checking equipment health, and ensuring compliance with safety standards. This can enhance operational efficiency and reduce the risk of equipment failures. Adaptation for underwater exploration and inspection of underwater structures, pipelines, or marine environments. Submersible rovers can contribute to the maintenance and monitoring of underwater infrastructure.

Conclusion

In conclusion, the development and implementation of a railway track inspection rover represent a significant stride towards revolutionizing the maintenance and monitoring of railway infrastructure. This project aimed to address the challenges associated with traditional manual inspections by introducing an autonomous and technologically advanced solution.

The railway track inspection rover, equipped with cutting-edge sensors, cameras, and navigational systems, demonstrated its effectiveness in providing real-time, accurate data about the condition of tracks. By autonomously traversing the railway network, the rover offered a comprehensive and detailed analysis of track geometry, alignment, and potential defects.

The project showcased the potential of this technology to enhance safety, reduce downtime, and optimize maintenance practices. The rover's ability to access challenging terrains, navigate through complex structures, and operate in various environmental conditions positions it as a valuable tool for ensuring the integrity and reliability of railway infrastructure.

In essence, the railway track inspection rover project underscores the transformative potential of autonomous technologies in the rail industry. The successful deployment of such rovers holds the promise of safer, more efficient, and sustainable railway operations, marking a significant step forward in the evolution of railway infrastructure maintenance.

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